$$v_\infty = \sqrt{\frac{2E_{k\infty}}{m}} = U\sqrt{\frac{2C_0}{\left(1-e^2\right)m}} \qquad (32)$$

The time interval between these collisions is

$$T = \frac{d}{\frac{v'_\infty + v_\infty}{2}} \qquad (33)$$

Hence the average current through the external circuit is

$$I = \frac{q}{T} \qquad (34)$$

From (31)、(32)、(33)、(34),

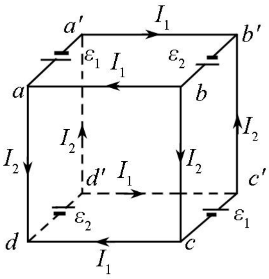
$$I = \frac{C_0 U^2}{d}\sqrt{\frac{1+eC_0}{1-e2m}} \qquad (35)$$

Points 1 8 (21)(28)(30)、(32)(29)(27) or : total; : ; : each;
2 (35) : .

VII.

Method 1:

1. The induced emfs in $a'a$ and cc' are



$$E_{a'a} = E_{cc'} = \varepsilon_1 = \frac{\sqrt{2}}{2}Bl^2\omega\sin\theta \qquad (1)$$

The induced emfs in $b'b$ and dd' are

$$\varepsilon_{b'b} = \varepsilon_{dd'} = \varepsilon_2 = \frac{\sqrt{2}}{2}Bl^2\omega\cos\theta \qquad (2)$$

Circuit symmetry gives

$$I_{ba} = I_{a'b'} = I_{d'c'} = I_{cd} = I_{1'},\quad I_{ad} = I_{d'a'} = I_{c'b'} = I_{bc} = I_2 \qquad (3)$$

Kirchhoff' s current law gives

$$I_{aa'} = I_{c'c} = I_1 - I_2 \qquad (4)$$

$$I_{b'b} = I_{dd'} = I_1 + I_2 \qquad (5)$$

Kirchhoff' s voltage law gives

$$I_1 R + I_{aa'}R + I_1 R_1 + I_{bb'}R = \varepsilon_2 - \varepsilon_1 \qquad (6)$$

$$I_1 R + I_{dd'}R + I_2 R - I_{ad}R = \varepsilon_2 + \varepsilon_1 \qquad (7)$$

$$I_{bc} = I_{ad} = I_1 = \frac{\sqrt{2}}{8}\frac{Bl^2\omega}{R}\left(\cos\theta - \sin\theta\right) \qquad (8)$$

$$I_{ad} = I_{bc} = I_2 = \frac{\sqrt{2}}{8}\frac{Bl^2\omega}{R}\left(\cos\theta + \sin\theta\right) \qquad (9)$$